A First Analysis of Feedback 2

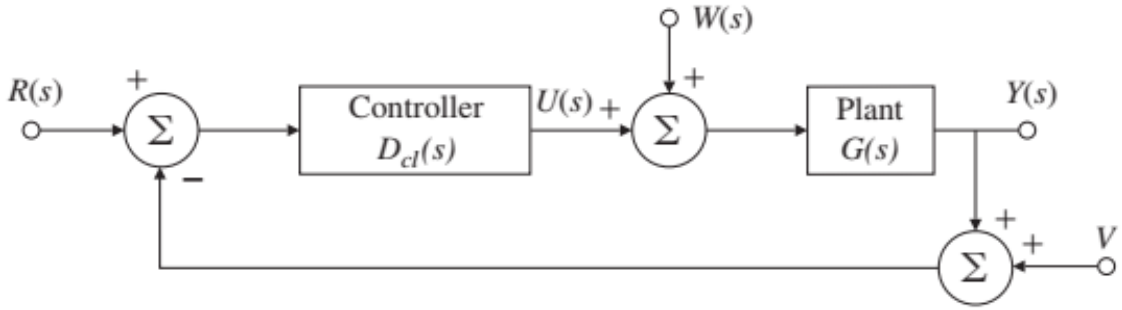
Professor: Juan David Bastidas Rodríguez

Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

Content

1. The three-term controller: PID control
2. Feedforward control by plant model inversion

1. The three-term controller: PID control



$$D_c(s) = k_P + \frac{k_I}{s} + k_D s$$

P: Proportional I: Integral D: Derivative

1. The three-term controller: PID control

Proportional Control (P)

- The control signal is linearly proportional to the system error (amplifier)
- k_P is called the proportional gain

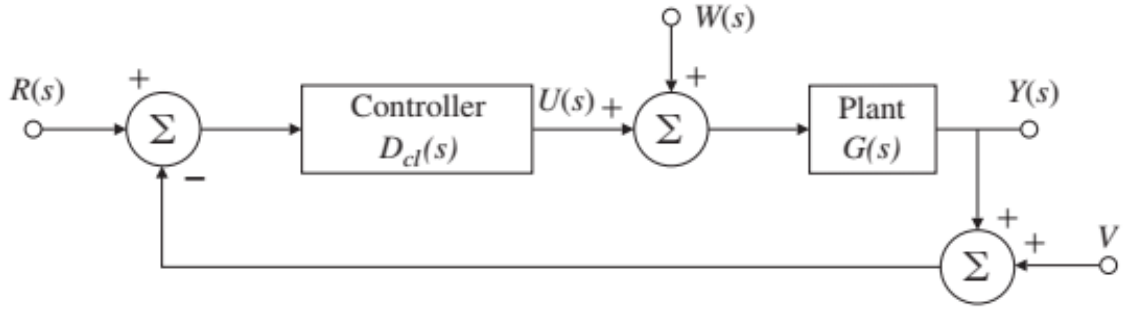
$$u(t) = k_P e(t),$$

$$\frac{U(s)}{E(s)} = D_{cl}(s) = k_P$$

1. The three-term controller: PID control

Proportional Control (P): for a 2nd order system

$$G(s) = \frac{A}{s^2 + a_1 s + a_2}$$ Plant



$$1 + k_P G(s) = 0$$ Characteristic equation

$$s^2 + a_1 s + a_2 + k_P A = 0$$

k_p increments produce
- Increment of w_n
- Reduction of ζ

$$w_n = \sqrt{a_2 + k_P A}$$

$$\zeta = \frac{a_1}{2w_n}$$

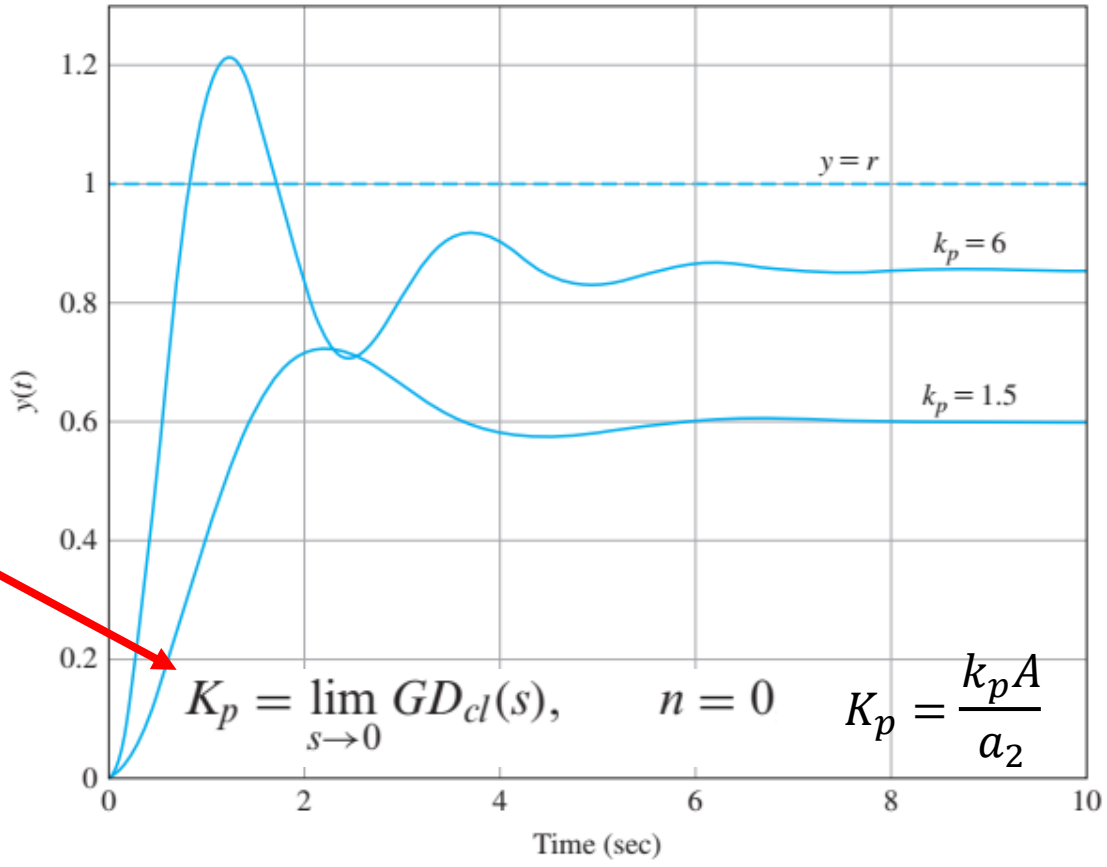
Closed-loop response with W(s) = 0 and V(s) = 0

$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{k_P G(s)}{1 + k_P G(s)}$$

1. The three-term controller: PID control

Proportional Control (P): for a 2nd order system

- The system is **type 0**
- e_{ss} due to a step in R(s) is constant $e_{ss} = 1/(1+\mathbf{K_p})$
- Making $a_1 = 1.4$, $a_2 = 1$, and $A = 1$
- y(t) for r = 1(t), with $k_P = 1.5$ and $k_P = 6$



1. The three-term controller: PID control

Proportional Control (P): for higher order systems

- k_P increments:
  - The damping of some of the poles might increase while decreasing in others
  - Increase the speed of response but larger transient overshoot
  - Lead to instability for a high enough gain
- Any **Type 0** system with **proportional control**
  - Will have a nonzero steady-state offset for constant reference input
  - Will not be capable of completely rejecting a constant disturbance input

1. The three-term controller: PID control

Integral Control (I): objectives

- To minimize the steady-state tracking error
- To minimize the steady-state error due to disturbances

- $\mathbf{k_I}$ is the integral gain
- u(t) is dynamic and depend on the error “history”

$$u(t) = k_I \int_{t_0}^{t} e(\tau)\, d\tau$$

$$\frac{U(s)}{E(s)} = D_{cl}(s) = \frac{k_I}{s}$$

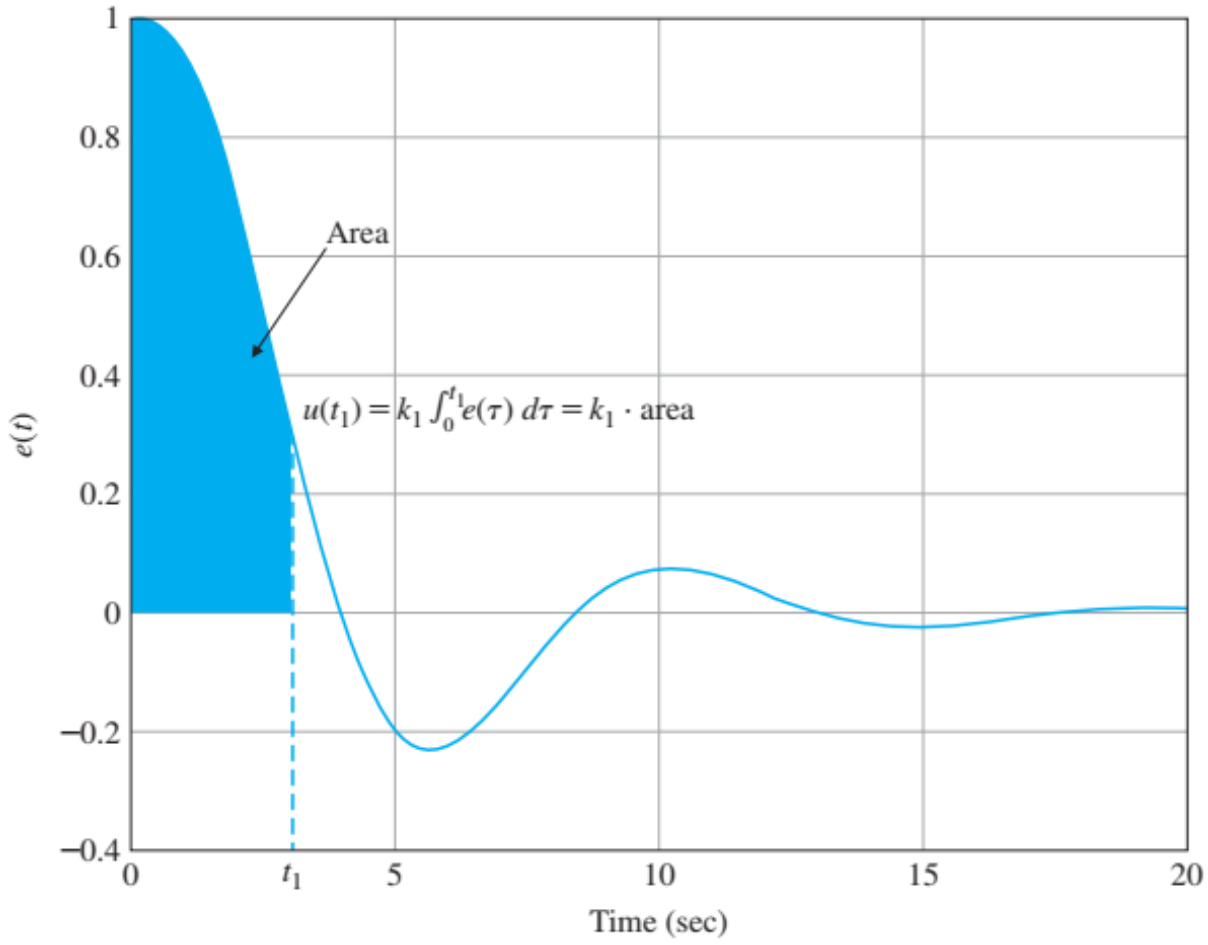
1. The three-term controller: PID control

Integral Control (I):

- The output is the integral of the error in a given **t**
- u(t) vary with t

$$u(t) = k_I \int_{t_0}^{t} e(\tau)\, d\tau$$

$$\frac{U(s)}{E(s)} = D_{cl}(s) = \frac{k_I}{s}$$

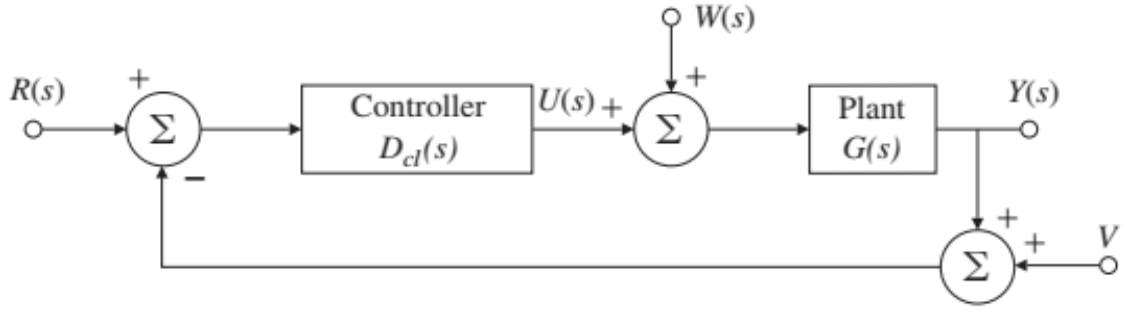


1. The three-term controller: PID control

Integral Control (I): for a 2nd order system

$$G(s) = \frac{A}{s^2 + a_1 s + a_2}$$ Plant

$$\frac{U(s)}{E(s)} = D_{cl}(s) = \frac{k_I}{s}$$ Controller



$$\frac{E(s)}{R(s)} = \frac{1}{1 + \frac{k_I}{s}G(s)} = \frac{s}{s + k_I G(s)}, \quad \frac{U(s)}{R(s)} = \frac{\frac{k_I}{s}}{1 + \frac{k_I}{s}G(s)} = \frac{k_I}{s + k_I G(s)}$$

$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{\frac{k_I}{s}G(s)}{1 + \frac{k_I}{s}G(s)} = \frac{k_I G(s)}{s + k_I G(s)}.$$

1. The three-term controller: PID control

Integral Control (I):

- With r(t) = 1(t) or R(s) = 1/s and using the Final Value Theorem
- Steady-state tracking error = 0, for **any k_I**
- **k_I** can be selected to provide an acceptable dynamic response
  - May lead to instability

Making $a_1 = 1.4$, $a_2 = 1$, and $A = 1$

$$y(\infty) = \frac{k_I G(0)}{0 + k_I G(0)} = 1, \quad e(\infty) = \frac{0}{0 + k_I G(0)} = 0,$$

$$u(\infty) = \frac{k_I}{0 + k_I G(0)} = G(0)^{-1} = 1.$$

From y(∞) it is clear that **k_I** should be:

- 1/G(0) or
- 1/DC gain of the plant

1. The three-term controller: PID control

Integral Control (I): output due to a step in W(s)

- Using the final value theorem

$$\frac{Y(s)}{W(s)} = \frac{sG(s)}{s + k_I G(s)}, \quad \frac{U(s)}{W(s)} = -\frac{k_I G(s)}{s + k_I G(s)}$$

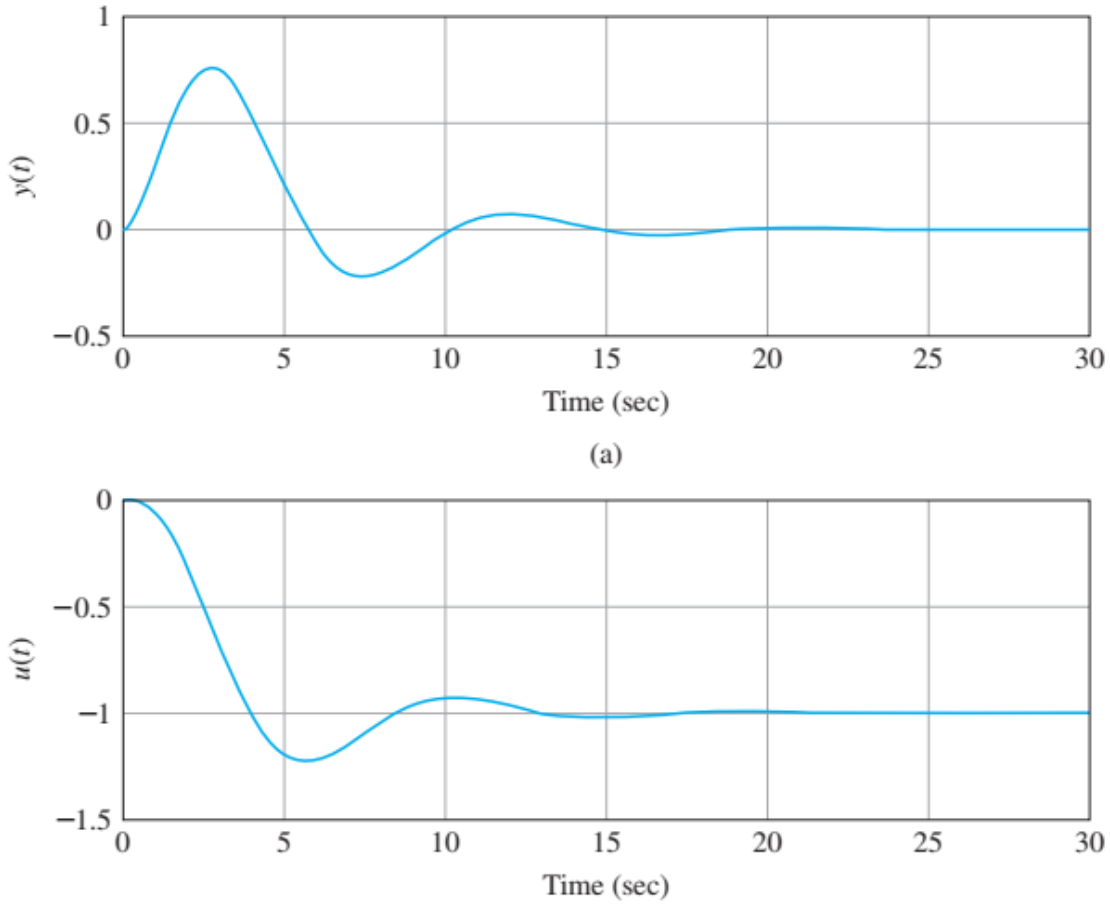
$$w(t) = 1(t) \qquad W(s) = 1/s.$$

$$y(\infty) = \frac{0 \cdot G(0)}{0 + k_I G(0)} = 0, \quad u(\infty) = -\frac{k_I G(0)}{0 + k_I G(0)} = -1$$

1. The three-term controller: PID control

Integral Control (I): output due to a step in W(s)

- Responses for $k_I = 0.5$
- The pole at s = 0 of the controller becomes the system **type 1**
- Zero steady-state output error
- This is **independent of the plant parameter values**
- This is **independent of the value of** k_I
- It is important to consider the actuator saturation: **anti-windup**



1. The three-term controller: PID control

Derivative Control (D): objectives

- To improve the closed-loop system stability
- To speed up the transient response
- To reduce the overshoot

  - $\mathbf{k_D}$ is the derivative gain
  - u(t) is dynamic and depend on the error derivative (rate of change)
  - u(t) = 0 if e(t) is constant
  - Tend to amplify the noise

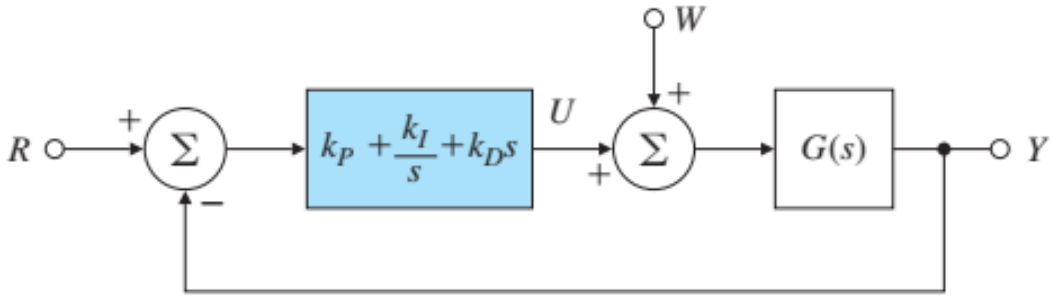
$$u(t) = k_D \dot{e}(t),$$

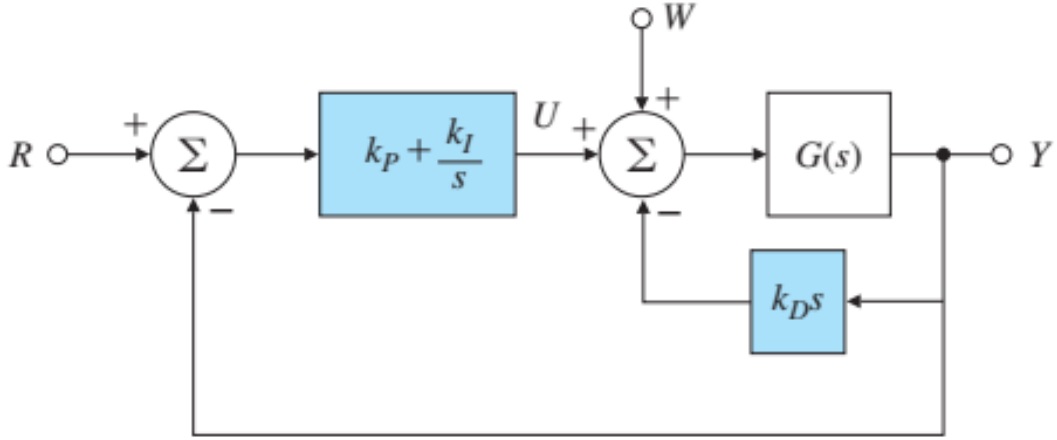
$$\frac{U(s)}{E(s)} = D_{cl}(s) = k_D s.$$

1. The three-term controller: PID control

Derivative Control (D): 2 implementation options

- Each option generate different zeros in the closed-loop TF





- Derivative of the error: control action response to changes in the **output (y(t)) and the input (r(t))**
- Derivative of the output: control action response to changes in the **output (y(t))**

1. The three-term controller: PID control

Proportional Plus Integral Control (PI):

- Allows for a faster response than a pure integral control
- The pole at s = 0 increments the Type of the system
  - The system can **reject completely constant bias disturbances**
- PID required for high-order systems

$$u(t) = k_P e(t) + k_I \int_{t_0}^{t} e(\tau)\, d\tau$$

$$\frac{U(s)}{E(s)} = D_{cl}(s) = k_P + \frac{k_I}{s}$$

1. The three-term controller: PID control

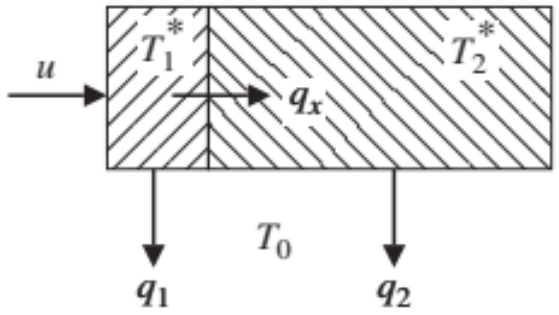
Proportional Plus Integral Control (PI): Example

PI Control of a Thermal System:

Consider the thermal system consisting of the lumped second-order model of two thermal masses connected by conduction described by G(s) with $\tau_1 = 1$, $\tau_2 = 10$, and $K_o = 1000$.

The goal is to **design a PI controller** to track the reference input temperature signal, r(t):

- Negligible overshoot
- Robustness with respect to perturbations in K_o, τ_1, and τ_2
- Explore the use of open-loop control, P control, and PI control



$$G(s) = \frac{K_o}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

$$\tau_2 = \frac{C_2}{H_x + H_1} \qquad \tau_1 = \frac{C_1}{H_x + H_1}$$

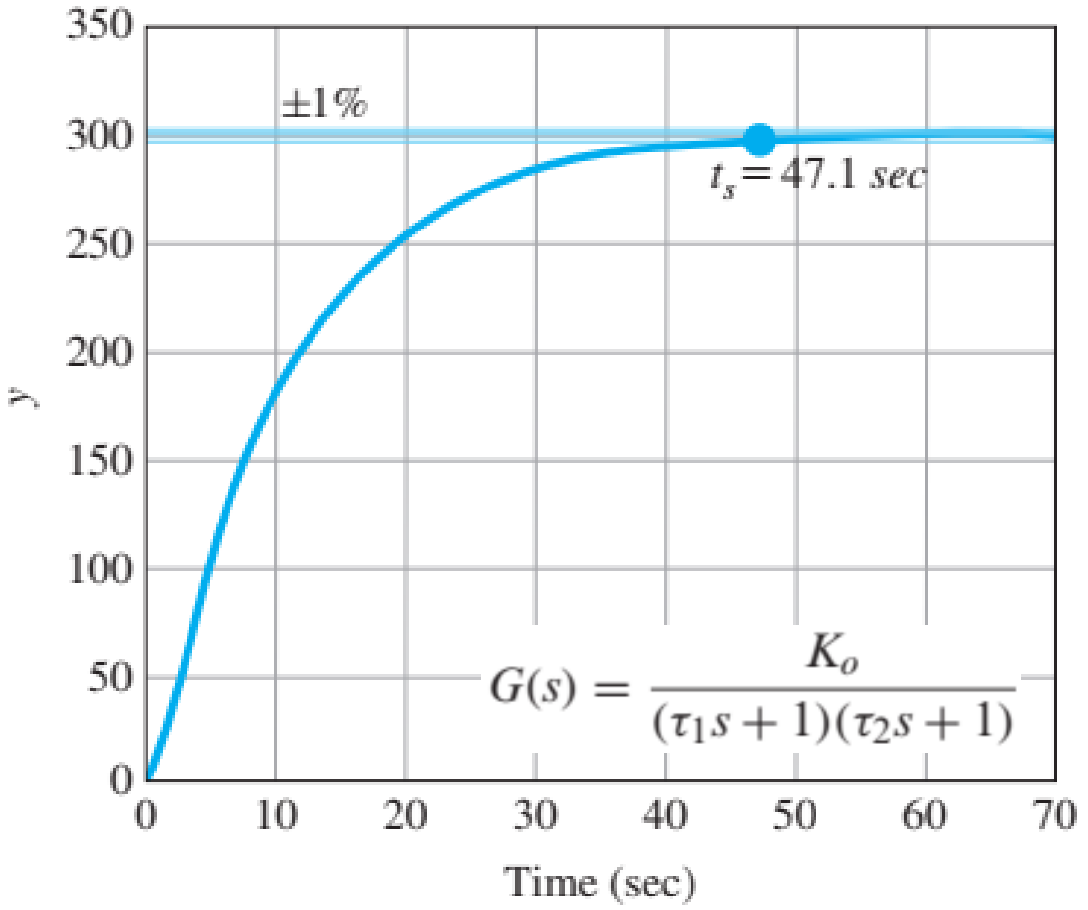
$$K_o = \frac{H_x}{(H_x + H_1)(H_x + H_2)}$$

1. The three-term controller: PID control

Proportional Plus Integral Control (PI): Example

PI Control of a Thermal System: **Open-loop control**

- Open-loop response for a step input of 0.3 (DC gain is 1000)
- Open-loop control:
  - High sensitivity to Ko
  - No feasible for tracking

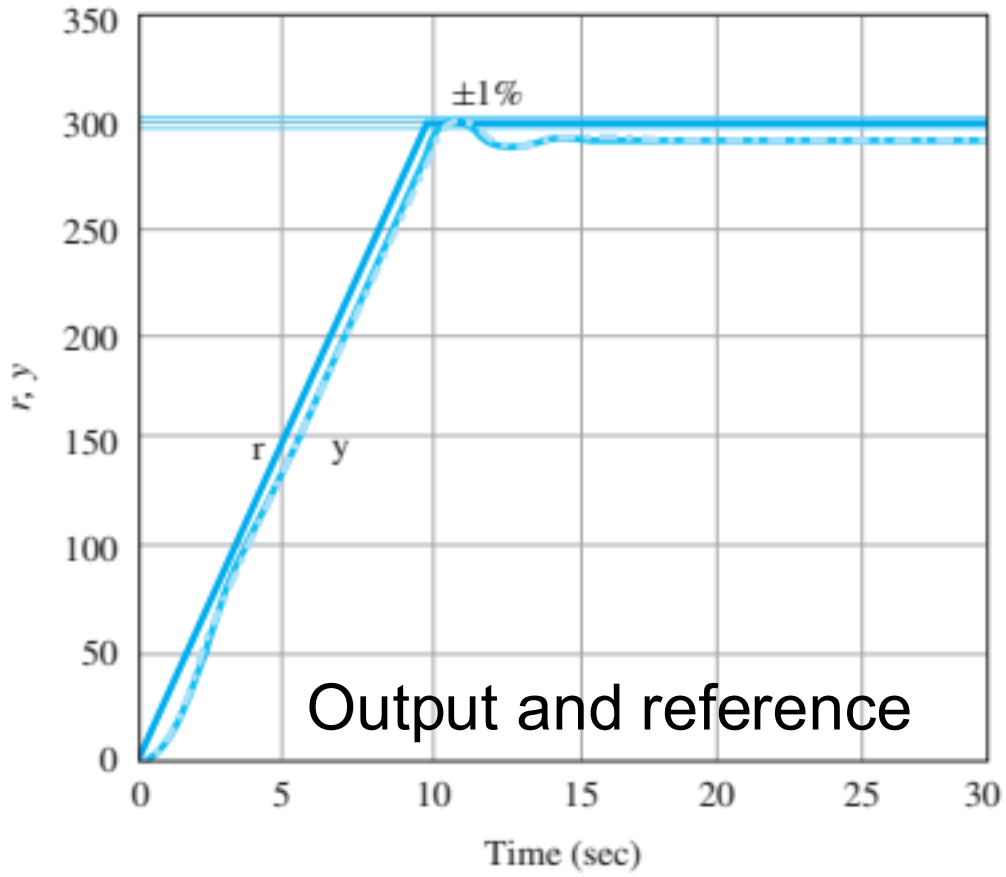


1. The three-term controller: PID control

Proportional Plus Integral Control (PI): Example

PI Control of a Thermal System: **P control (closed-loop)**

- For kp = 0.03 ζ = 0.3
- e_{ss} = 10 °C (not acceptable in some applications)
- ts ≈ 14 s (faster than OL)
- ±10% of K_o does not affect the response

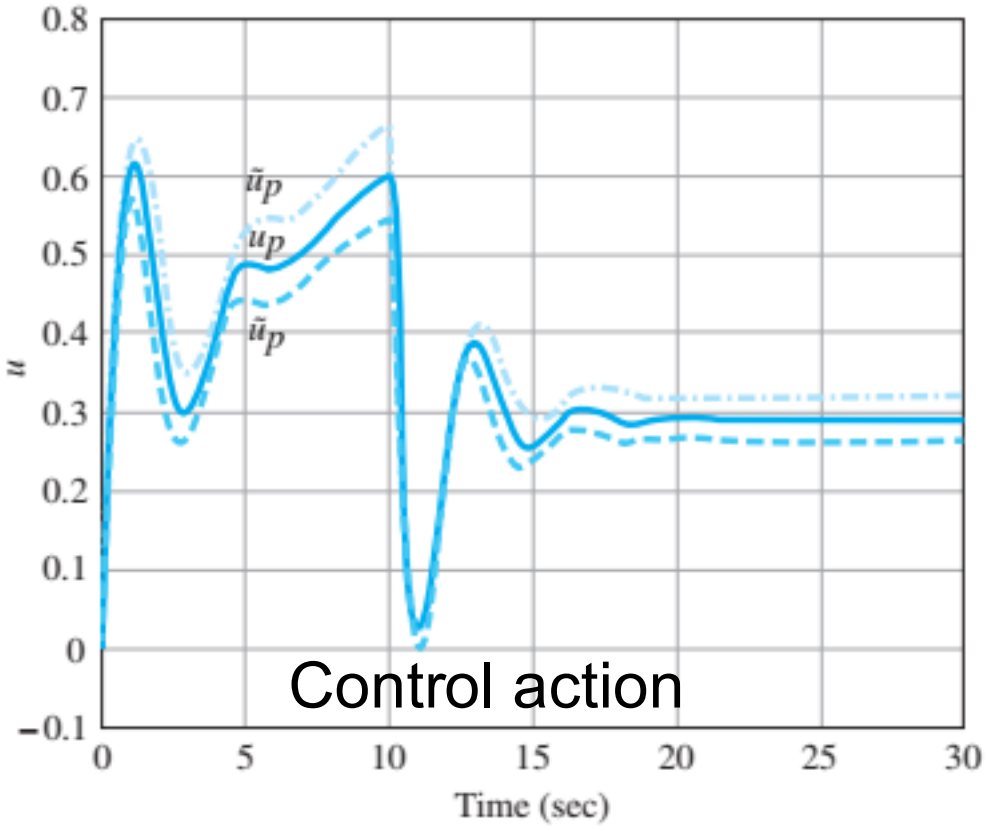


1. The three-term controller: PID control

Proportional Plus Integral Control (PI): Example

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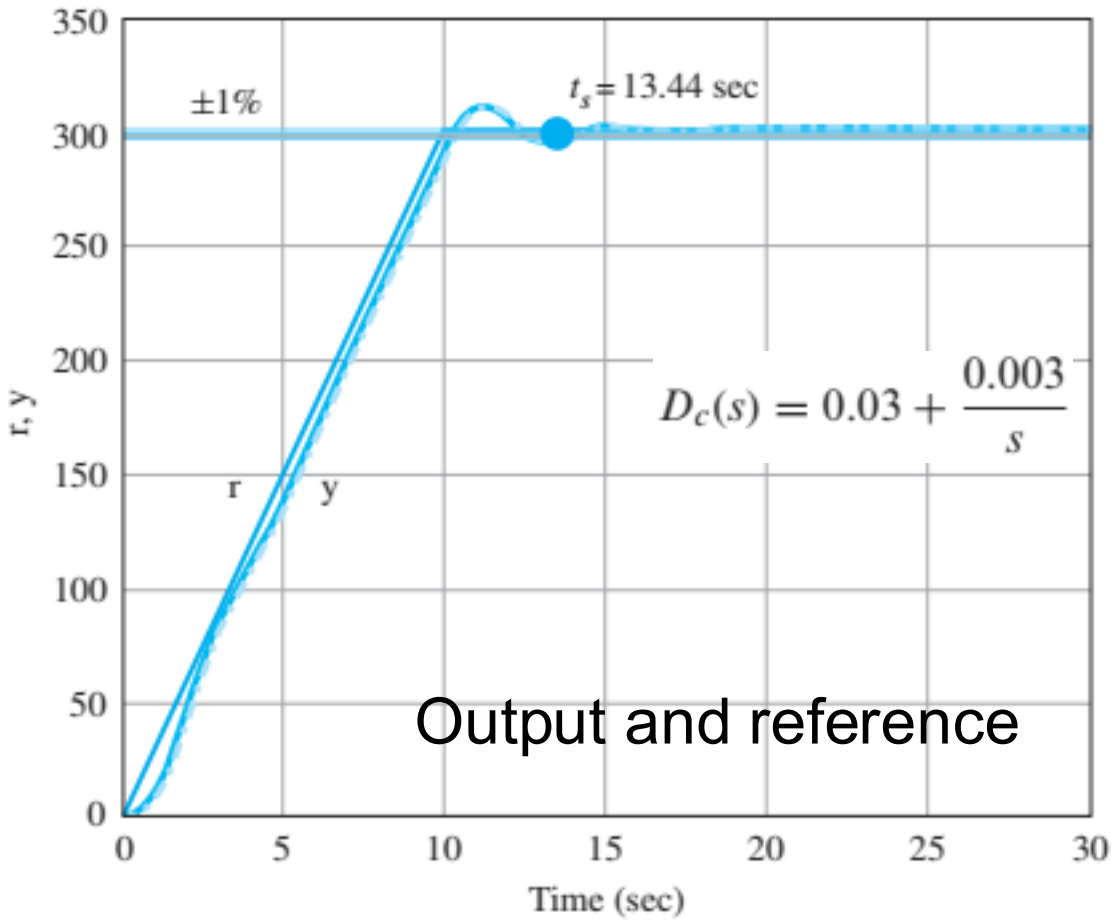


1. The three-term controller: PID control

Proportional Plus Integral Control (PI): Example

PI Control of a Thermal System: **PI control (closed-loop)**

- $k_P = 0.03$, $k_I = 0.003$
- $e_{ss} = 0$ °C
- ts = 13.44 s (faster than OL)
- ±10% of K_o does not affect the response
- There is an overshoot
- Zero at -0.1 of the PI cancels the pole of the plant at -0.1

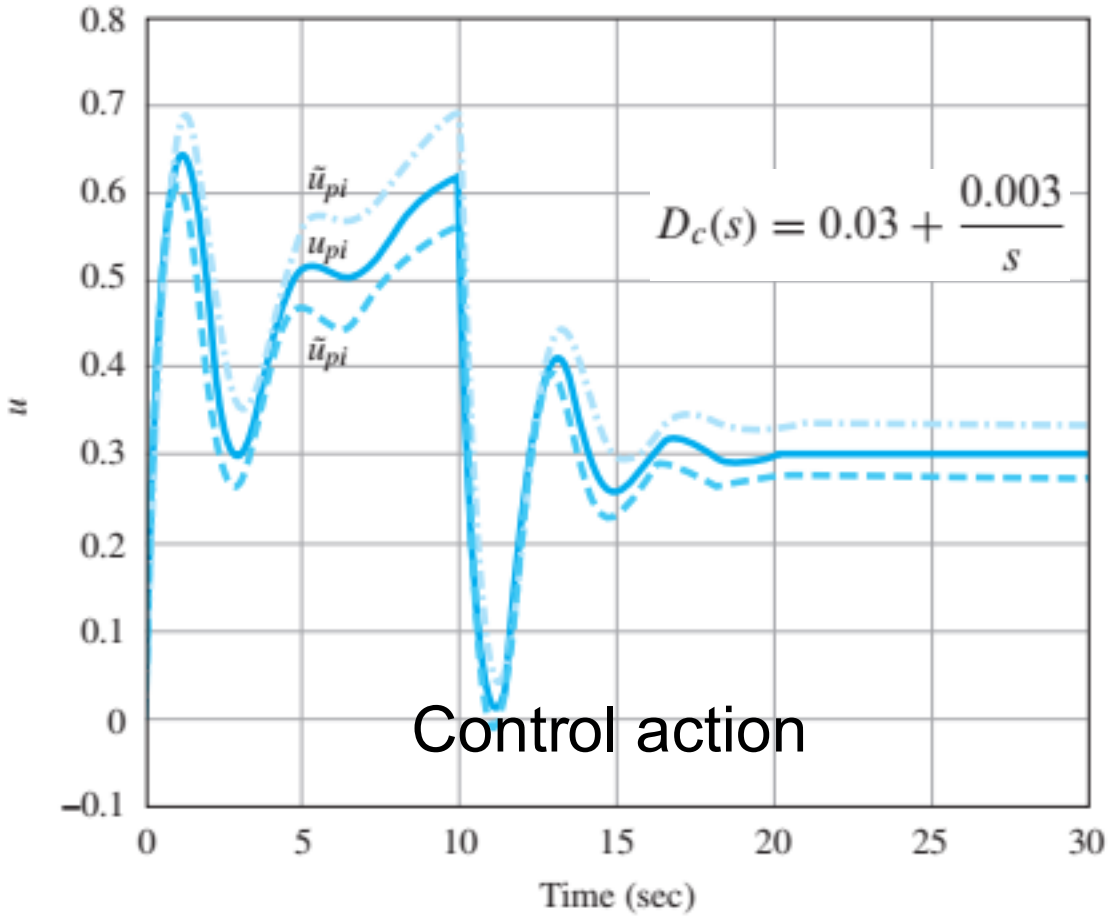


1. The three-term controller: PID control

Proportional Plus Integral Control (PI): Example

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1. The three-term controller: PID control

PID Control: Proportional, Integral, Derivative

$$u(t) = k_P e(t) + k_I \int_{t_0}^{t} e(\tau)\, d\tau + k_D \dot{e}(t), \qquad \frac{U(s)}{E(s)} = D_{cl}(s) = k_P + \frac{k_I}{s} + k_D s.$$

For a 2nd order plant $G(s) = \dfrac{A}{s^2 + a_1 s + a_2}$

Characteristic equation

$$s^2 + a_1 s + a_2 + A(k_P + \frac{k_I}{s} + k_D s) = 0,$$

$$s^3 + a_1 s^2 + a_2 s + A(k_P s + k_I + k_D s^2) = 0.$$

$$s^3 + (a_1 + A k_D) s^2 + (a_2 + A k_P) s + A k_I = 0.$$

The PID parameters can “**force**” the roots location of the closed-loop system

1. The three-term controller: PID control

PID Control: Example – DC motor speed control

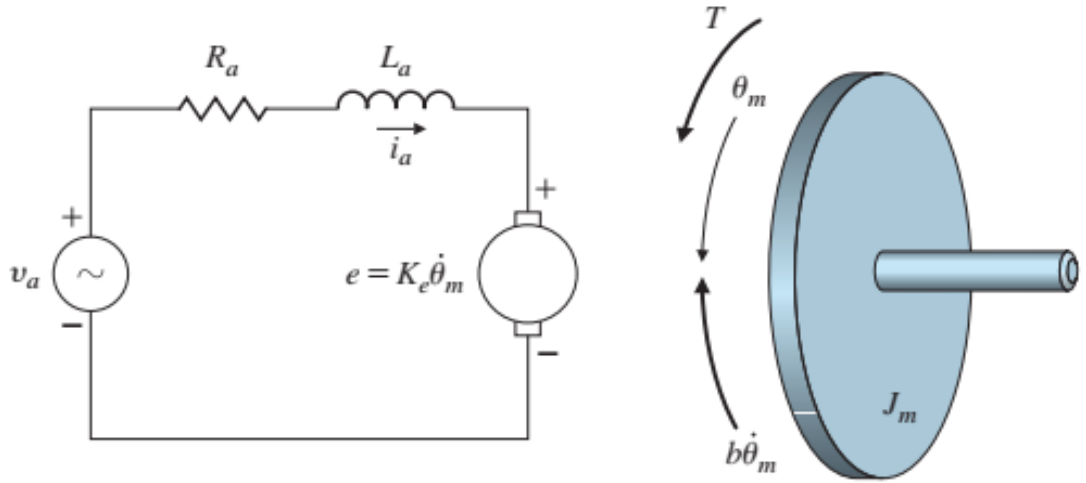
Consider the DC motor speed control with parameters

- $J_m = 1.13 \times 10^{-2}$ N·m· sec^2/rad
- b = 0.028 N·m·sec/rad
- $L_a = 10^{-1}$ H
- $R_a = 0.45\ \Omega$
- $K_t = 0.067$ N·m/amp
- $K_e = 0.067$ V·sec/rad

PID parameters (**different from the book**)

- $k_P = 2.609$
- $k_I = 6.259$ 1/sec
- $k_D = 0.2718$ sec

Analyze the responses of this system to steps in a disturbance torque and steps in the reference input using the three different controllers: P, PI, and PID

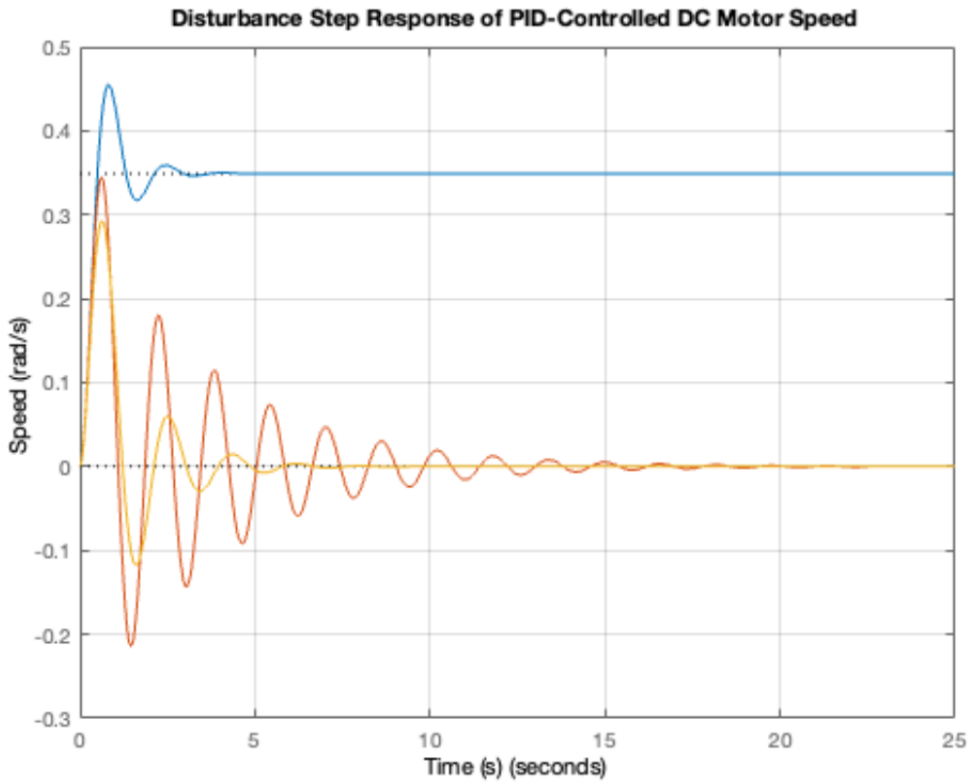


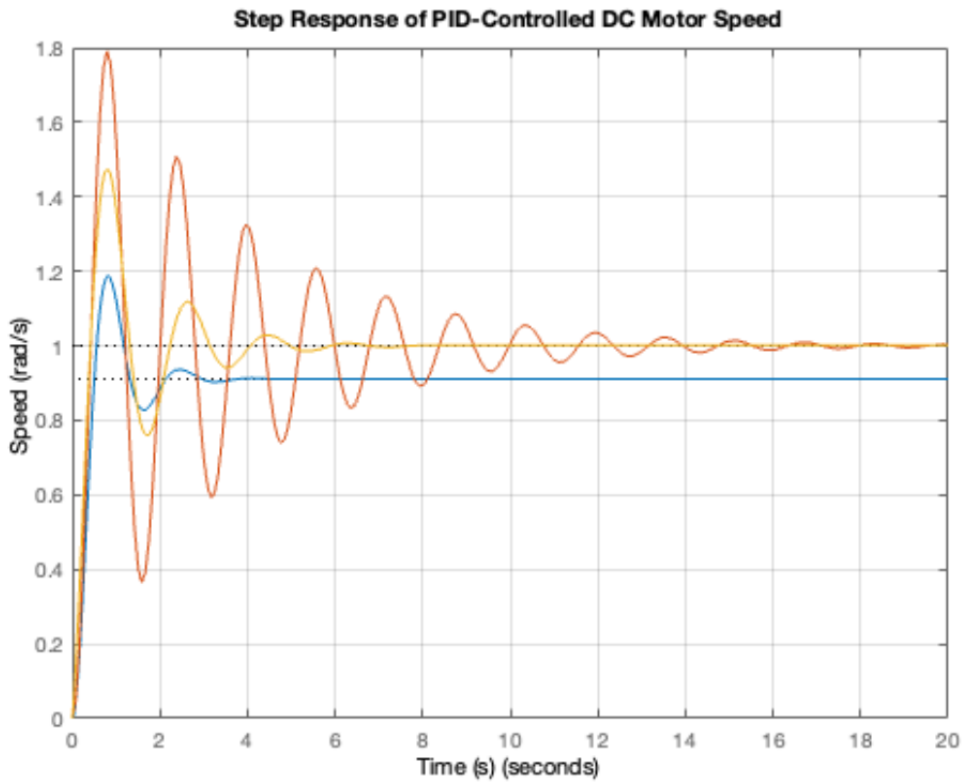
$$\frac{\Theta_m(s)}{V_a(s)} = \frac{K_t}{s\left[(J_m s + b)(L_a s + R_a) + K_t K_e\right]}$$

$$\frac{\Omega(s)}{V_a(s)} = s\frac{\Theta_m(s)}{V_a(s)}$$

1. The three-term controller: PID control

PID Control: Example – DC motor speed control





1. The three-term controller: PID control

PID Control: Example

```
clear; clc; close all;
% Motor parameters
Jm = 1.13e-2; % N·m· sec2/rad
b = 0.028; % N·m·sec/rad
La = 10e-1; % H
Ra = 0.45; % Ω
Kt = 0.067; % N·m/amp
Ke = 0.067; % V·sec/rad
% PID parameters
% In the book kp = 3, kI = 15 sec (error because kI is in 1/s) and
% kD = 0.3 sec, but they don't work. I calculated the parameters
by using
% sisotool and the sesion is seved in
"Ej_4_6_Sisotool_PID_Motor_Speed"
% tf(C) shows the TF of the controller in a simple way to define
the
% parameters
kP = 2.609;
kI = 6.259;
kD = 0.2718;
% Plant TF
s=tf('s');
Teta_Va = Kt/( s*((Jm*s+b)*(La*s+Ra)+Kt*Ke) );
Omega_Va = s*Teta_Va;
% Simplify plant
Omega_Va = minreal(Omega_Va);
% PID TF
% Just for P
Dcl1 = kP;
% For PI
Dcl2 = kP + kI/s;
% For PID
Dcl3 = kP + kI/s + kD*s;
%% Calculate the response for a step in the input
% Close-loop TF
Tcl1 = feedback(Omega_Va*Dcl1,1);
Tcl2 = feedback(Omega_Va*Dcl2,1);
Tcl3 = feedback(Omega_Va*Dcl3,1);
% Simplify closed-loop TF
Tcl1 = minreal(Tcl1);
Tcl2 = minreal(Tcl2);
Tcl3 = minreal(Tcl3);
% Plot the results: Step response in the input
figure(1);
hold on;
step(Tcl1);
step(Tcl2);
step(Tcl3);
hold off;
title('Step Response of PID-Controlled DC Motor Speed');
ylabel('Speed (rad/s)');
xlabel('Time (s)');
grid on;
%% Calculate the response for a step in the Disturbance
% Close-loop TF
Tw1 = feedback(Omega_Va,Dcl1);
Tw2 = feedback(Omega_Va,Dcl2);
Tw3 = feedback(Omega_Va,Dcl3);

% Simplify closed-loop TF
Tw1 = minreal(Tw1);
Tw2 = minreal(Tw2);
Tw3 = minreal(Tw3);

% Plot the results: Step response in the input
figure(2);
hold on;
step(Tw1);
step(Tw2);
step(Tw3);
hold off;

title('Disturbance Step Response of PID-Controlled DC Motor
Speed');
ylabel('Speed (rad/s)');
xlabel('Time (s)');
grid on;
```

1. The three-term controller: PID control

PID Control: Example – DC-DC converter

Consider the control of a DC-DC converter using the unity feedback structure. Assume the inductor L and capacitor C constitute the output filter for the converter while the converter input voltage is Vin and the load is Rc.

(a) Use the proportional controller

Dc(s) = k_P, and determine the **system type** and **steady-state error** properties with respect to **disturbance inputs**.

(b) Let the control be PI as given by Dc(s) = $k_P + k_I/s$ and determine the **system type** and **steady-state error** properties with respect for **disturbance inputs**.

$$G(s) = V_{in}\frac{1}{LCs^2 + \frac{L}{R_c}s + 1}$$

1. The three-term controller: PID control

PID Control: Example – DC-DC converter

a)

$$T_w(s) = -\frac{V_{in}}{\left(LCs^2 + \frac{L}{R_c}s + 1\right) + V_{in}K_p}$$

$$= s^0 T_{o,w},$$

$$n = 0,$$

$$K_{o,w} = -\frac{(1 + V_{in}K_p)}{V_{in}}.$$

Applying the equation

$$y_{ss} = \lim_{s\to 0}\left[sT_w(s)\frac{1}{s^{k+1}}\right]$$

$$= \lim_{s\to 0}\left[T_{o,w}(s)\frac{s^n}{s^k}\right].$$

The system is **Type 0** and the steady-state error to a **unit-step disturbance** is:

$$e_{ss} = -\frac{V_{in}}{1 + V_{in}k_p}$$

1. The three-term controller: PID control

PID Control: Example – DC-DC converter

b)

$$T_w(s) = -\frac{V_{in}s}{s\left(LCs^2 + \frac{L}{R_c}s + 1\right) + V_{in}(k_p s + k_I)},$$

$$n = 1,$$

$$K_{1,w} = -k_I,$$

Applying the equation

$$y_{ss} = \lim_{s\to 0}\left[sT_w(s)\frac{1}{s^{k+1}}\right]$$

$$= \lim_{s\to 0}\left[T_{o,w}(s)\frac{s^n}{s^k}\right].$$

The system is **Type 1** and the steady-state error to a **unit-ramp disturbance** is:

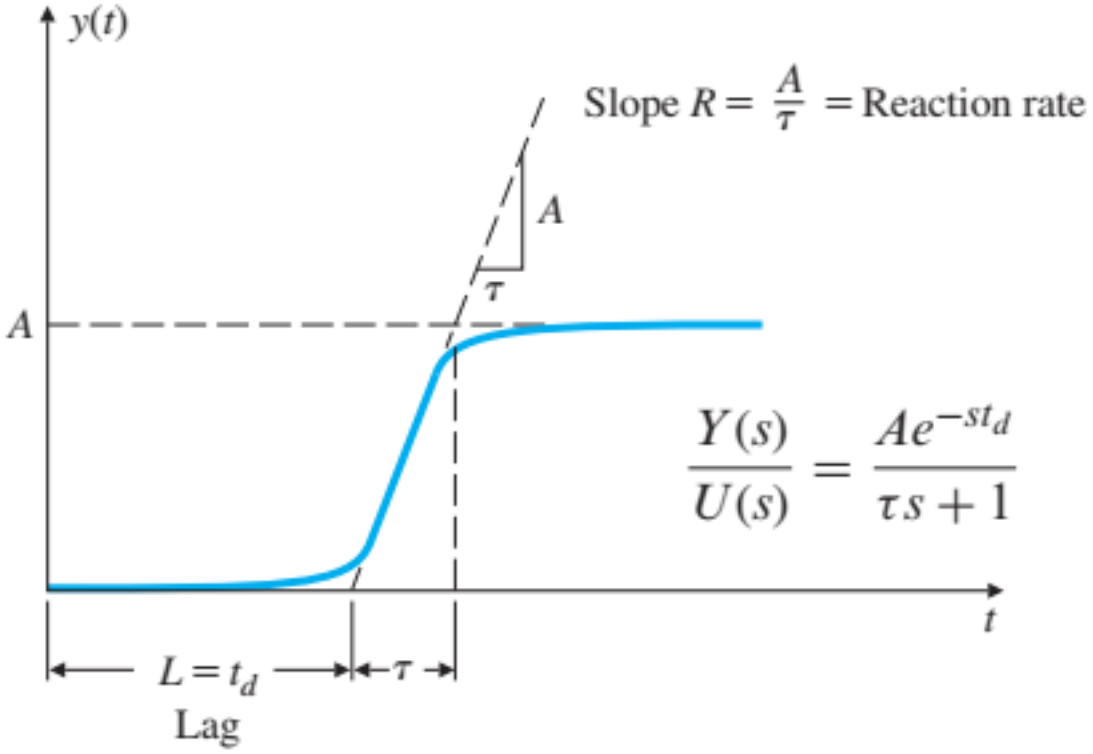
$$e_{ss} = -\frac{1}{k_I}$$

1. The three-term controller: PID control

Ziegler–Nichols Tuning of the PID Controller

- 1st-order system with a time delay or “transportation lag” of td sec
- Can be obtained from unit-step response
- 2 methods to calculate PID parameters
  - ζ = 0.21 (from Mp to 0.25 Mp in 1 period)
  - From marginal stability condition
- **Process operator perform a final tuning**

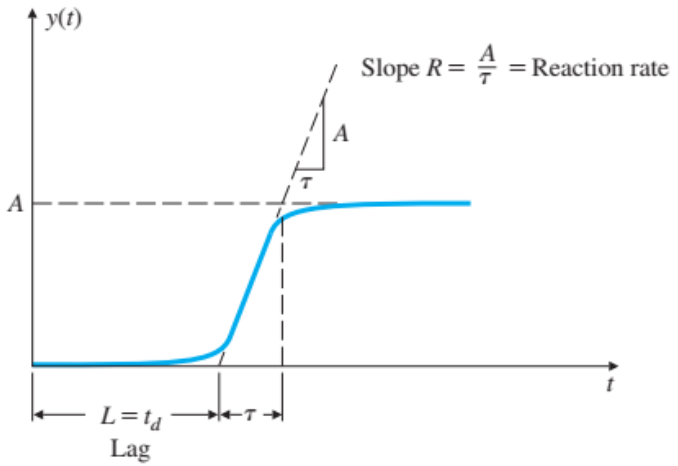
Process reaction curve of many systems

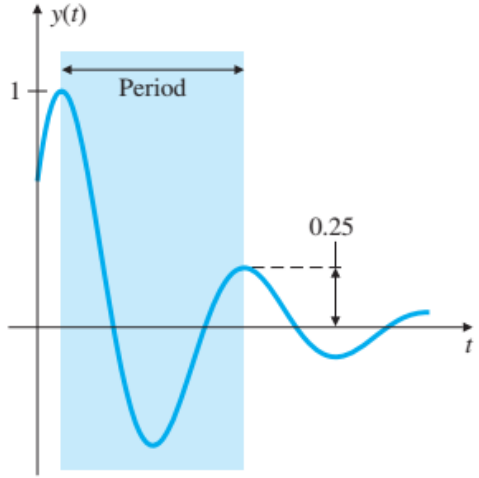


1. The three-term controller: PID control

Ziegler–Nichols Tuning of the PID Controller

Method 1 ($\zeta = 0.21$):





$$D_c(s) = k_P\left(1 + \frac{1}{T_I s} + T_D s\right)$$

Ziegler–Nichols Tuning for the Regulator
$D_c(s) = k_P(1 + 1/T_I s + T_D s)$, for a Decay Ratio of 0.25

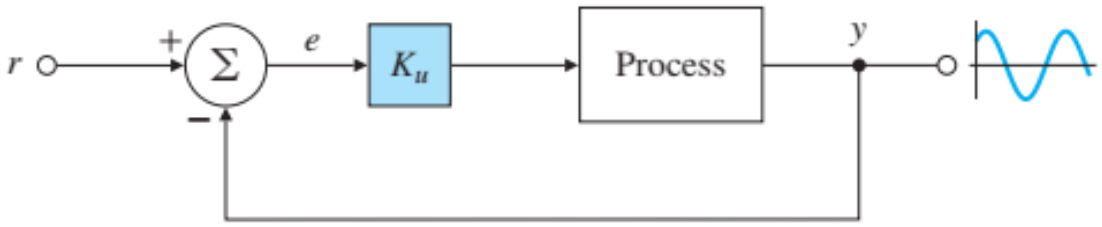
Type of Controller	Optimum Gain
P	$k_P = 1/RL$
PI	$k_P = 0.9/RL$, $T_I = L/0.3$
PID	$k_P = 1.2/RL$, $T_I = 2L$, $T_D = 0.5L$

1. The three-term controller: PID control

Ziegler–Nichols Tuning of the PID Controller

Method 2 (Marginal stability):

- Ku: K value that makes y(t) sinusoidal
- Pu: Period of y(t)



$$D_c(s) = k_P\left(1 + \frac{1}{T_I s} + T_D s\right)$$

Ziegler-Nichols Tuning for the Regulator $D_c(s) = k_P(1 + 1/T_I s + T_D s)$, Based on the Ultimate Sensitivity Method

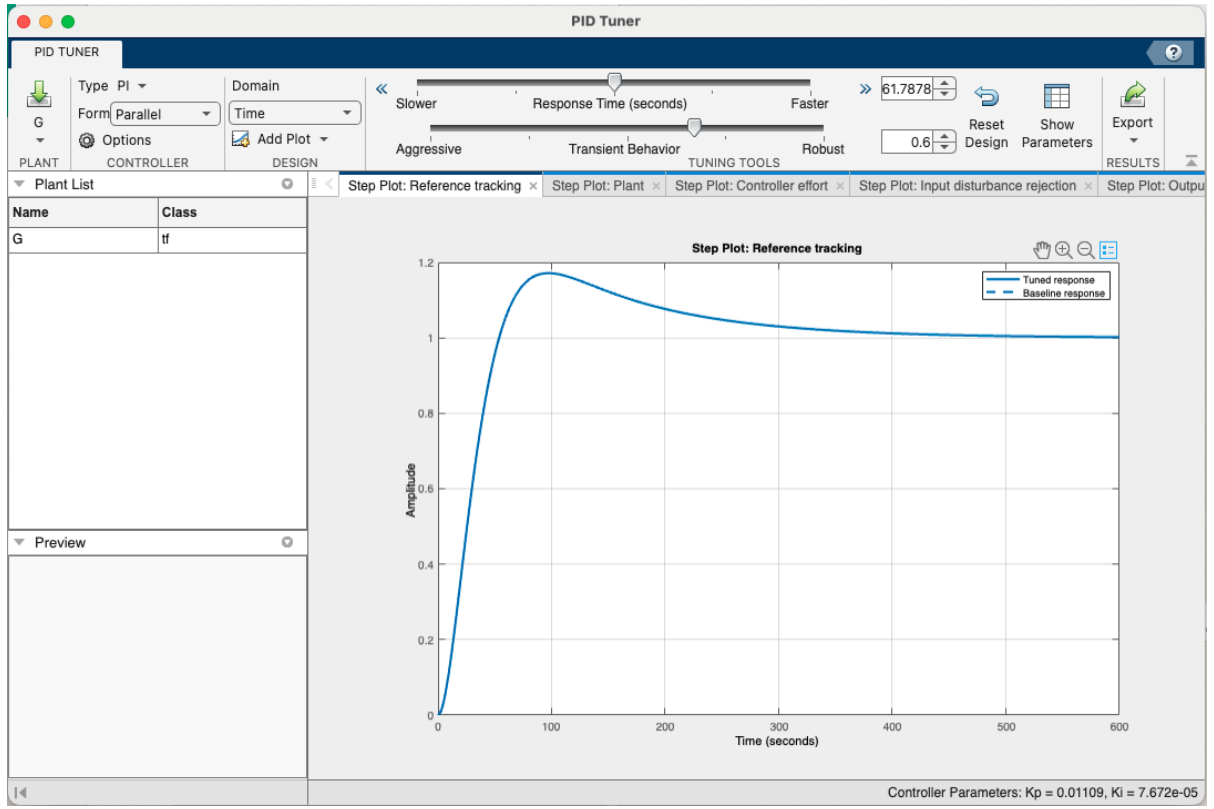
Type of Controller	Optimum Gain
P	$k_P = 0.5K_u$
PI	$k_P = 0.45K_u$, $T_I = \frac{P_u}{1.2}$
PID	$k_P = 1.6K_u$, $T_I = 0.5P_u$, $T_D = 0.125P_u$

1. The three-term controller: PID control

Tuning of the PID Controller with Matlab: pidTuner

Three-fold objectives:

- Stability
- Performance
- Robustness

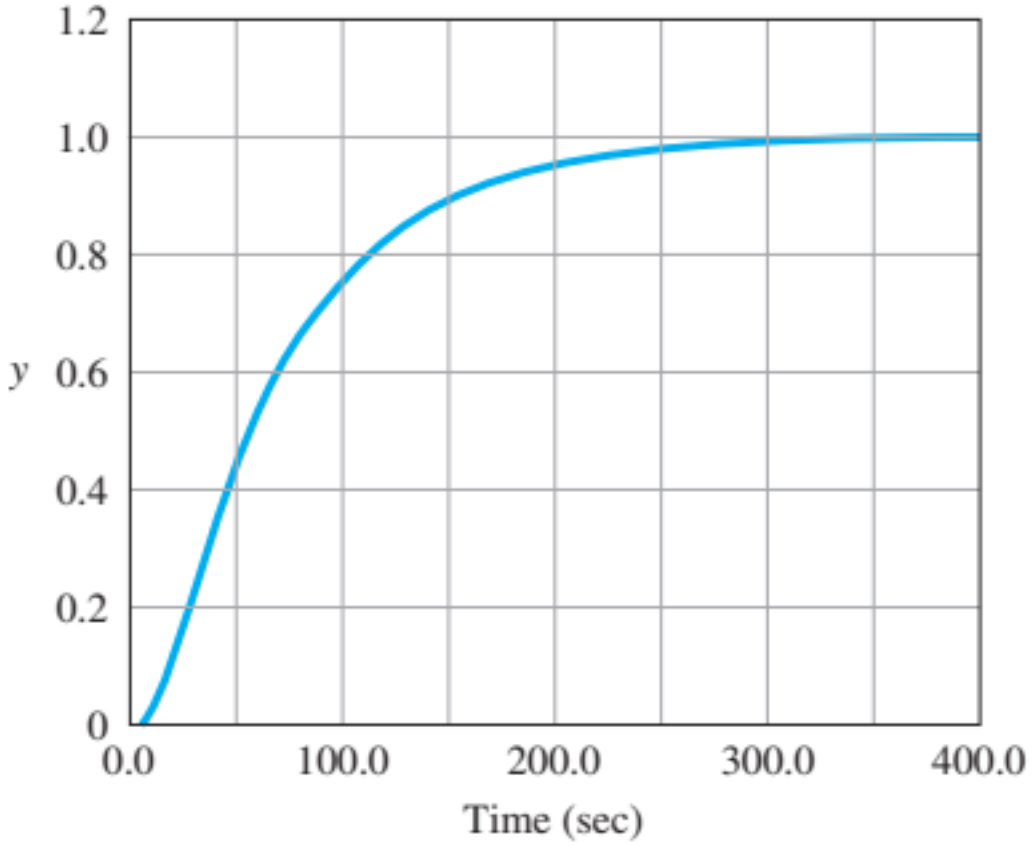



```
G = tf(0.3,[1,0.1,0]); % plant model
C = pidtune(G,'PI'); % design a PI for G
pidTuner(G,C); % pidTuner app
```

1. The three-term controller: PID control

Tuning of the PID Controller: Example

Tuning of a Heat Exchanger (method 1 - $\zeta = 0.21$): Determine proportional and PI regulator gains for the system using the Ziegler–Nichols

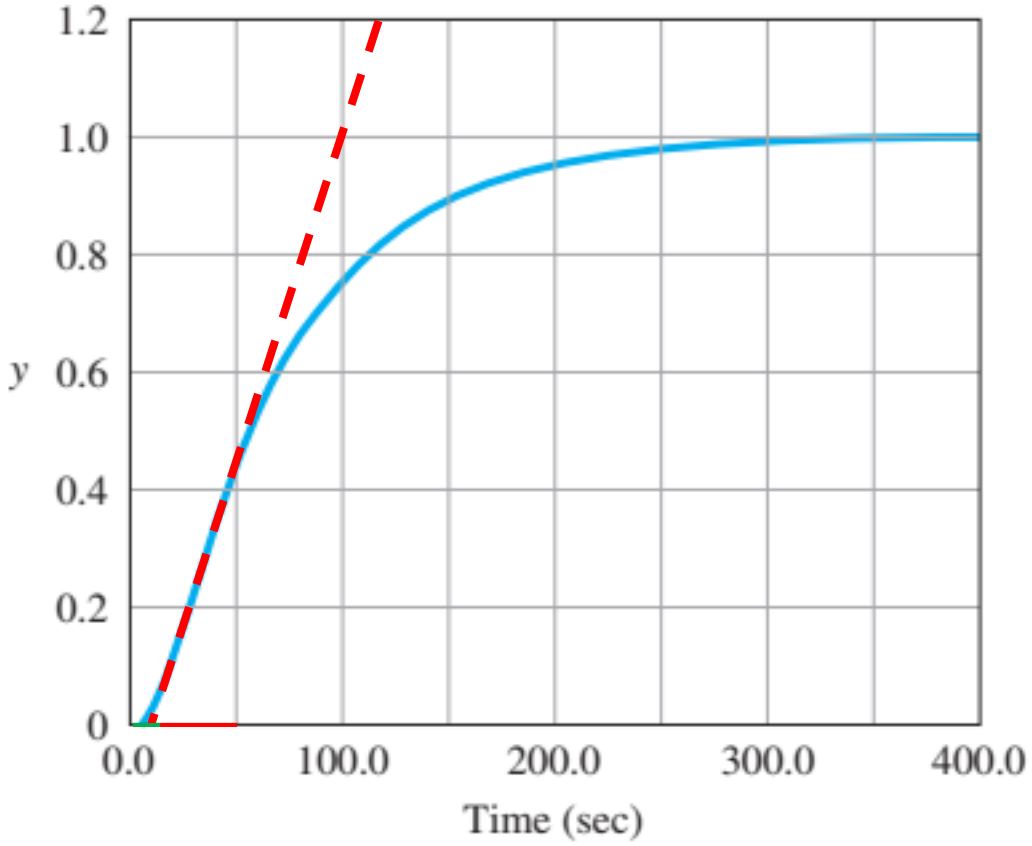


1. The three-term controller: PID control

Tuning of the PID Controller: Example (method 1 - ζ = 0.21)

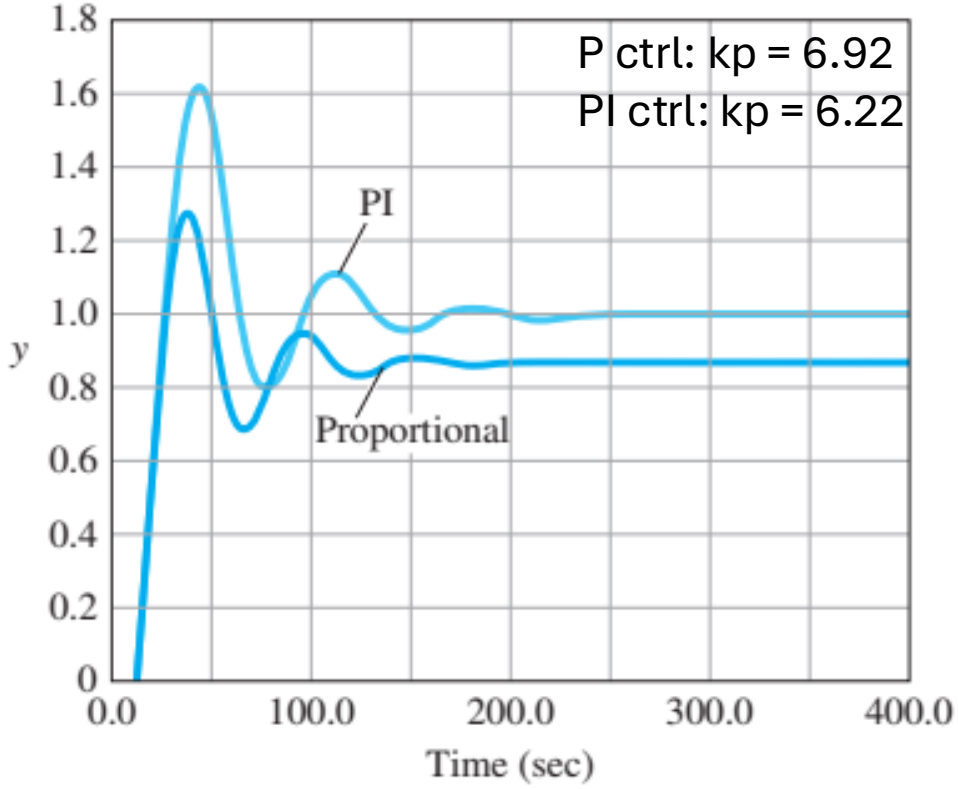
- R ≅ A/τ = 1/90 (slope)
- L ≅ 13 sec (delay)
- P ctrl: $k_P = \frac{1}{RL} = \frac{90}{13} = 6.92$
- PI ctrl: $k_P = \frac{0.9}{RL} = 6.22$

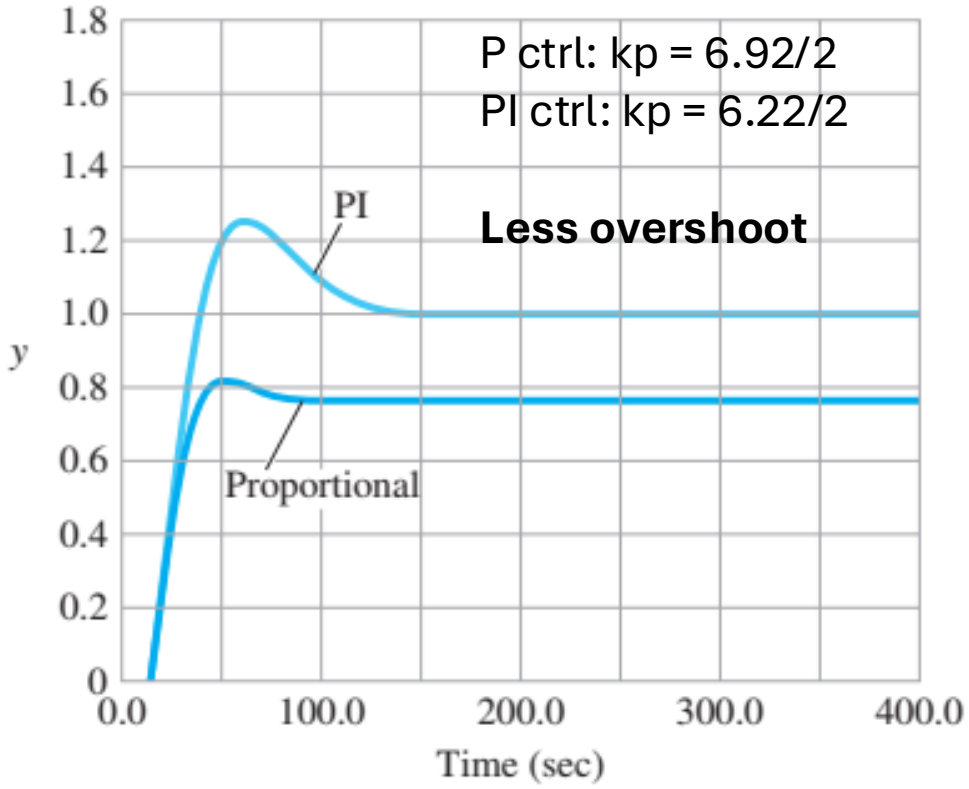
$$T_I = \frac{L}{0.3} = \frac{13}{0.3} = 43.3$$



1. The three-term controller: PID control

Tuning of the PID Controller: Example (method 1 - ζ = 0.21)





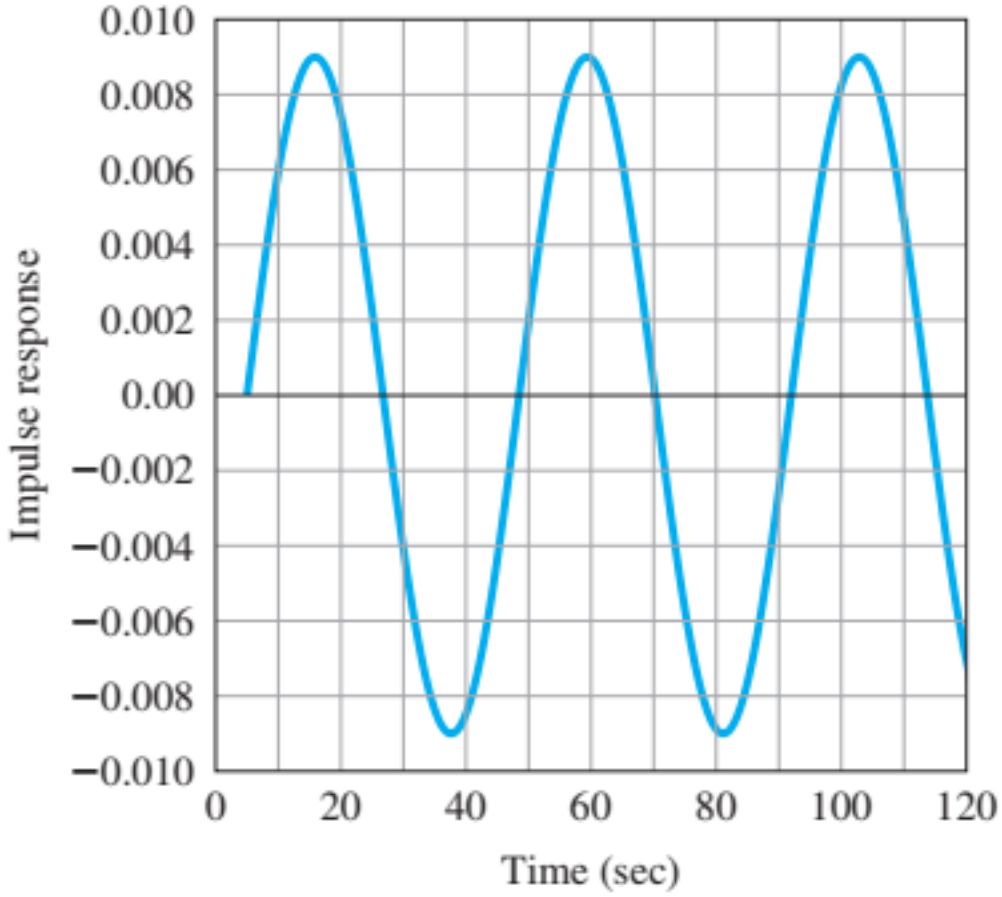
1. The three-term controller: PID control

Tuning of the PID Controller: Example

Tuning of a Heat Exchanger (method 2 – marginal stability)

- kp of a P controller was increased until obtain this response for kp = 15.3
- Pu ≅ 42 sec

Determine the P and PI controllers.



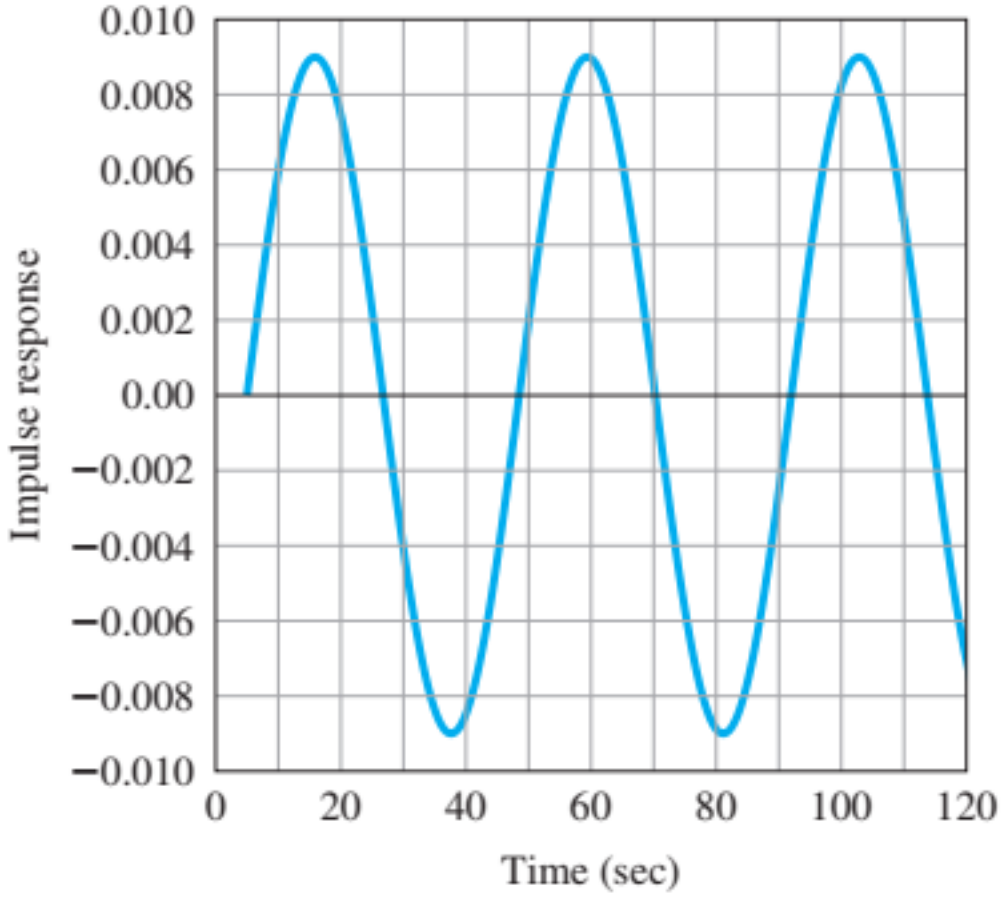
1. The three-term controller: PID control

Tuning of the PID Controller: Example (method 2 – marginal stability)

- P ctrl: $k_P = 0.5K_u, \quad k_P = 7.65$

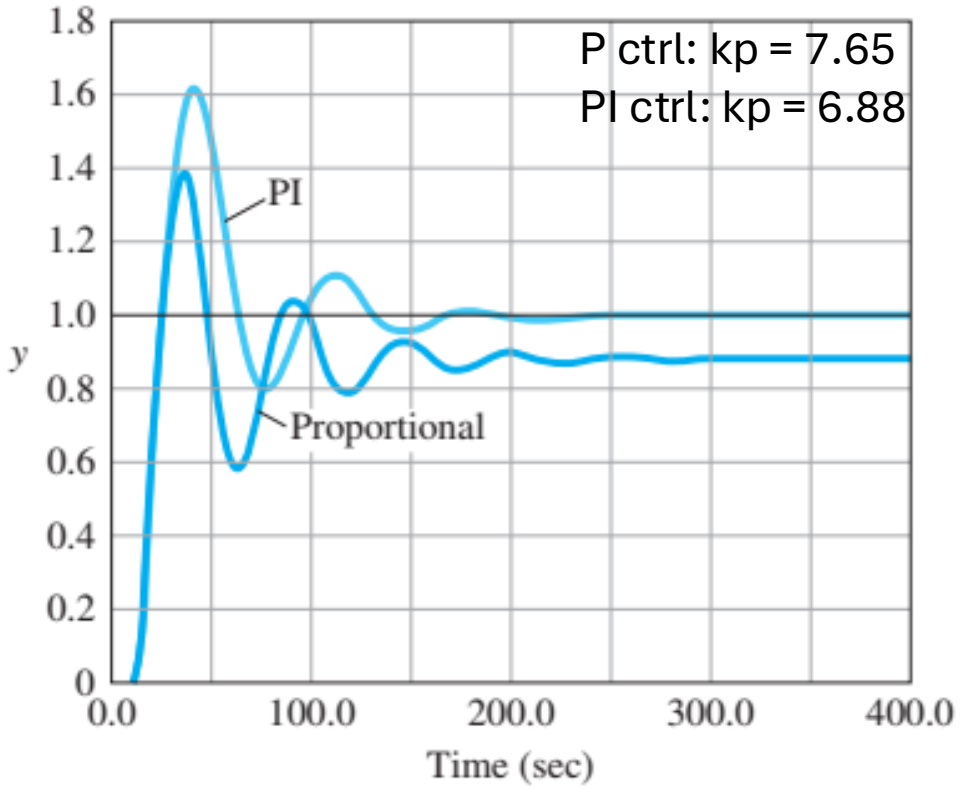
- PI ctrl: $k_P = 0.45K_u, \quad k_P = 6.885,$

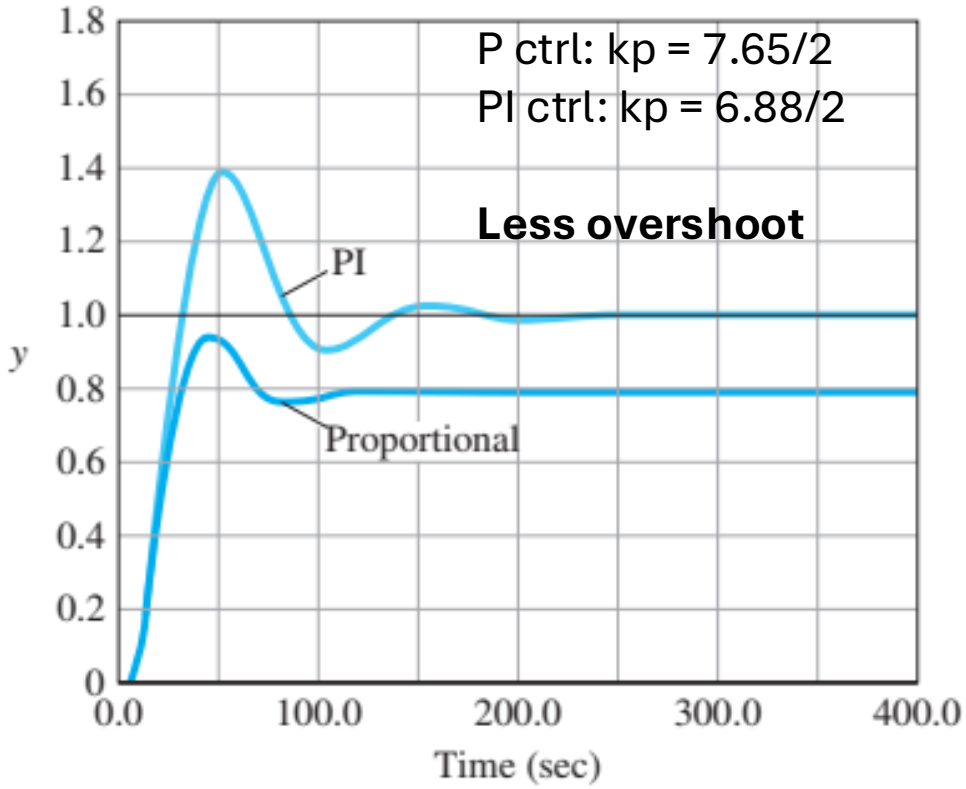
$$T_I = \frac{1}{1.2}P_u = 35$$



1. The three-term controller: PID control

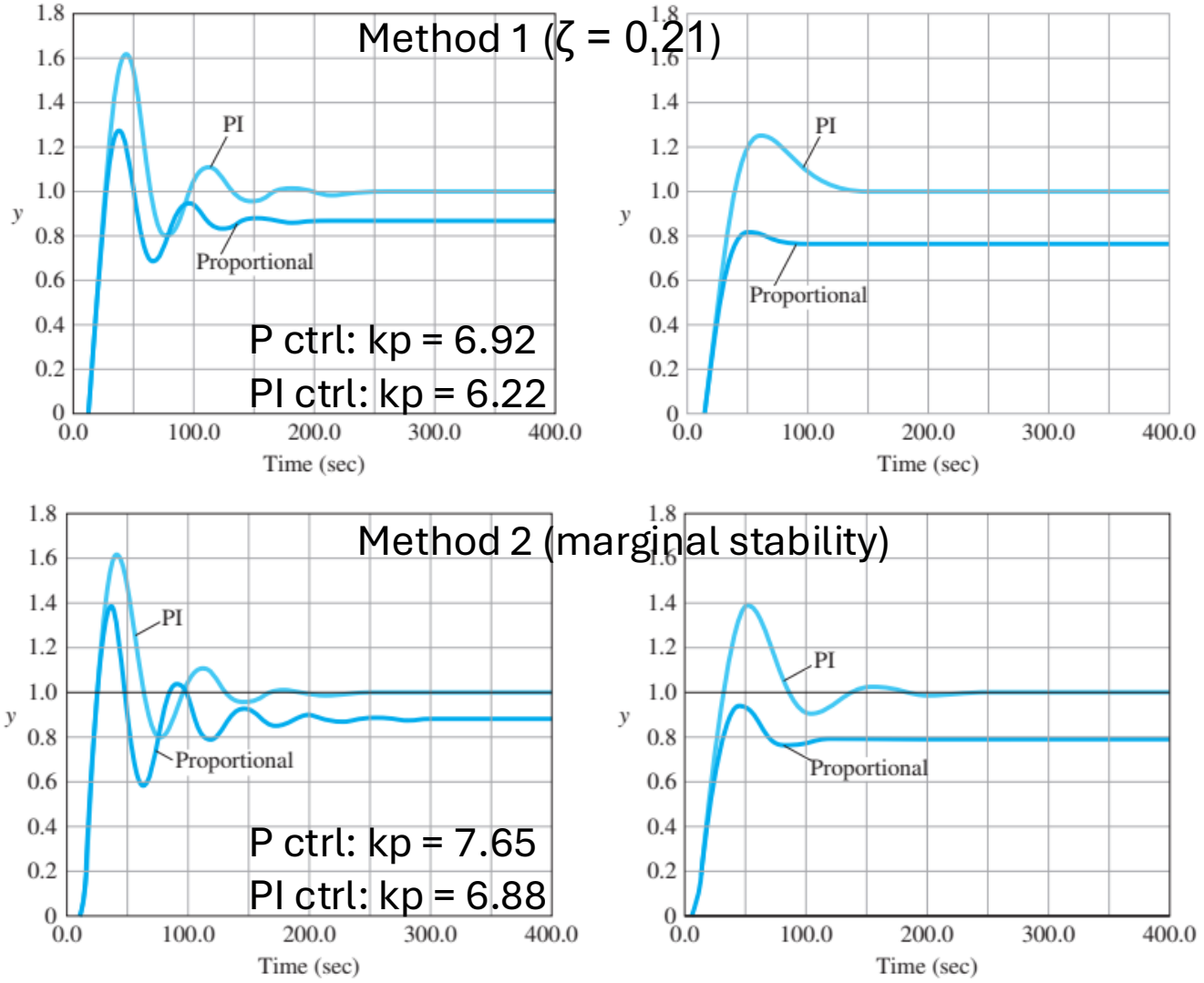
Tuning of the PID Controller: Example (method 2)





1. The three-term controller: PID control

Tuning of the PID Controller: Example
comparison of methods 1 and 2



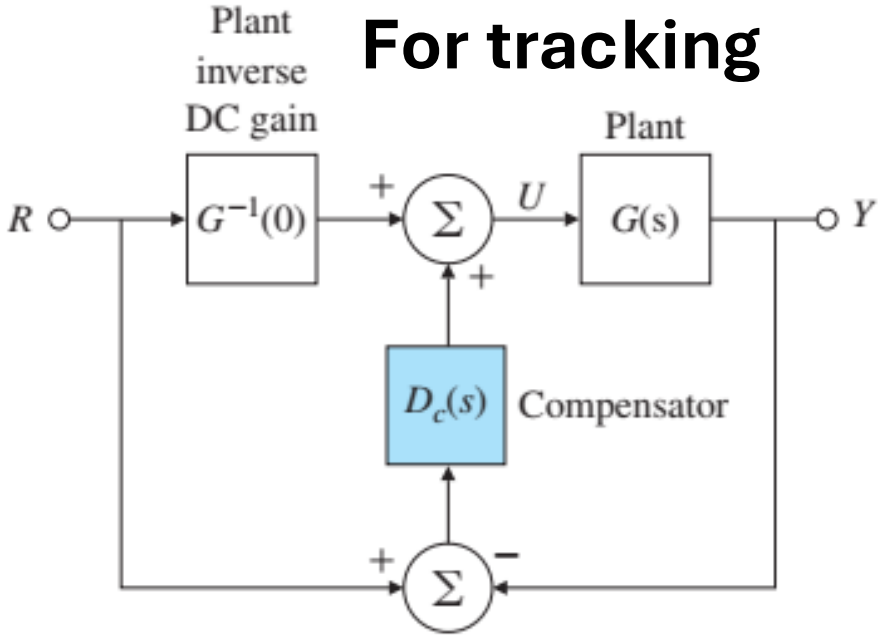
2. Feedforward Control by Plant Model Inversion

- **Feedforward:** provides the control effort required for the desired command input
- **Feedback:** takes care of the differences between the real plant and the plant model plus the effects of any disturbances.

2. Feedforward Control by Plant Model Inversion

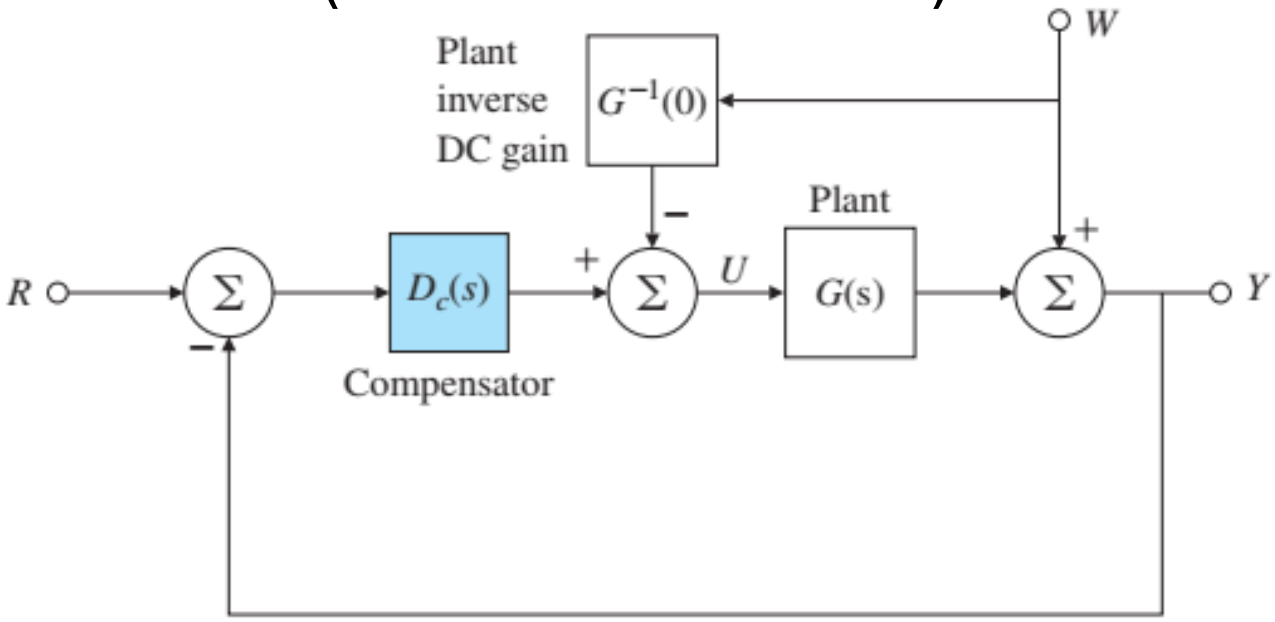
More complex feedforward control can be used by inverting G(s) over an entire frequency range.

For tracking



For disturbance rejection

(if W is measurable)



2. Feedforward Control by Plant Model Inversion

Example: Feedforward Control for DC Motor

- G(s) is the TF of the plant with a1 = 1.4, a2 = 1, and A = 1
- Proportional controller with kp = 1.5;

(a) Use feedforward control to eliminate the steady-state tracking error for a step reference input.

(b) Also use feedforward control to eliminate the effect of a constant output disturbance signal on the output of the system.

$$G(s) = \frac{A}{s^2 + a_1 s + a_2}$$

2. Feedforward Control by Plant Model Inversion

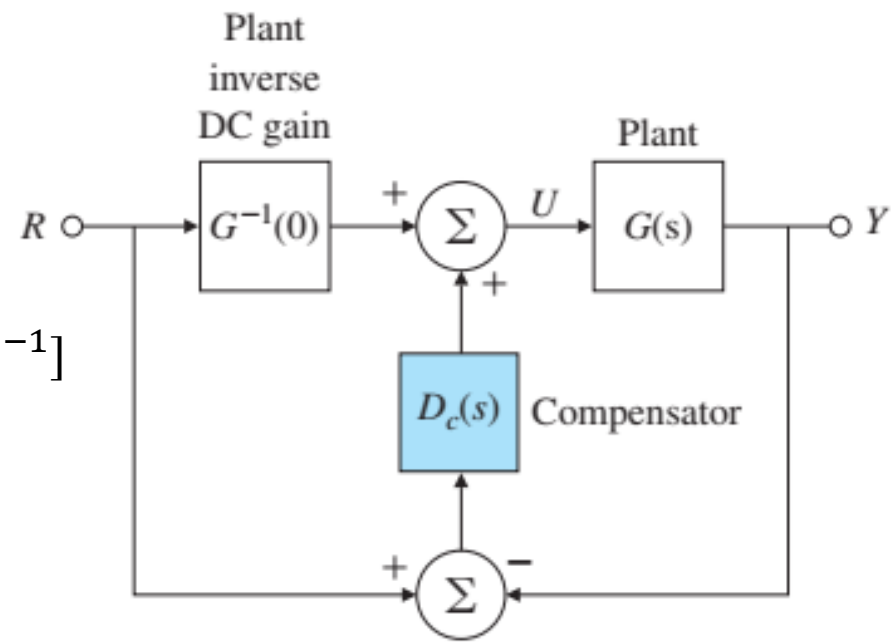
Example: Feedforward Control for DC Motor

(a) Tracking

- DC gain is $G^{-1}(0) = 1$
- Dc(s) = kP

$$G(s) = \frac{A}{s^2 + a_1 s + a_2}$$

Structure for tracking



Closed-loop TF:

$$Y(s) = G(s)[k_P E(s) + R(s)\,G(0)^{-1}]$$

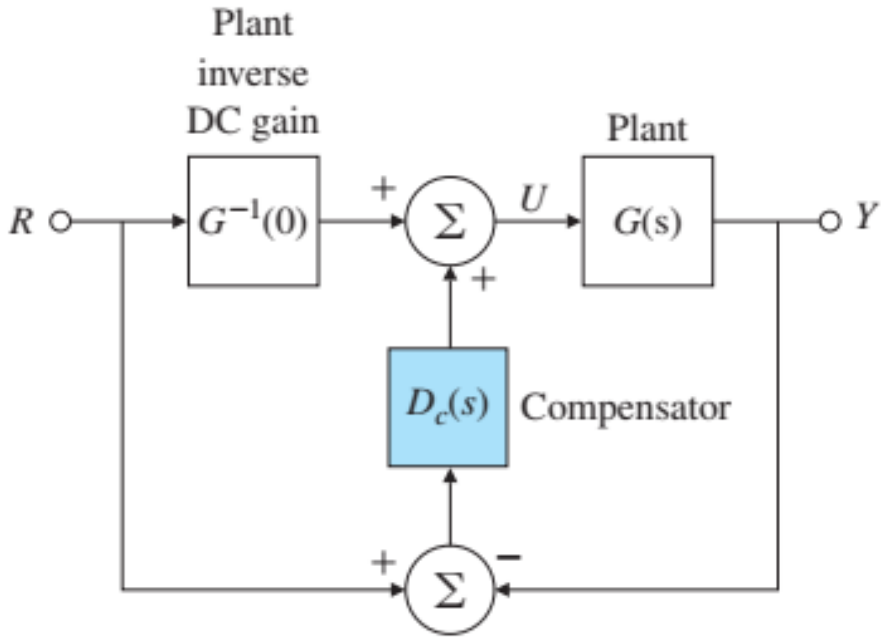
$$E(s) = R(s) - Y(s),$$

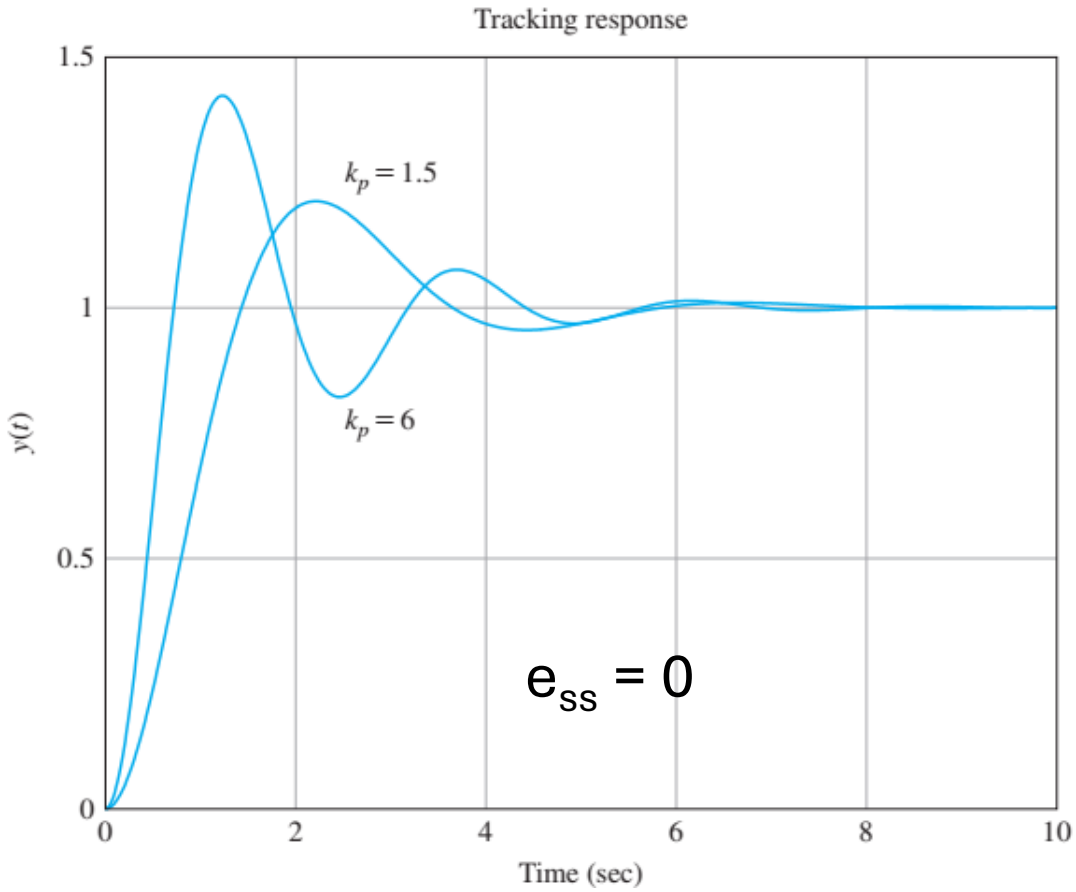
$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{k_P G(s) + G(0)^{-1}}{1 + k_P G(s)}$$

2. Feedforward Control by Plant Model Inversion

Example: Feedforward Control for DC Motor

(a) Tracking





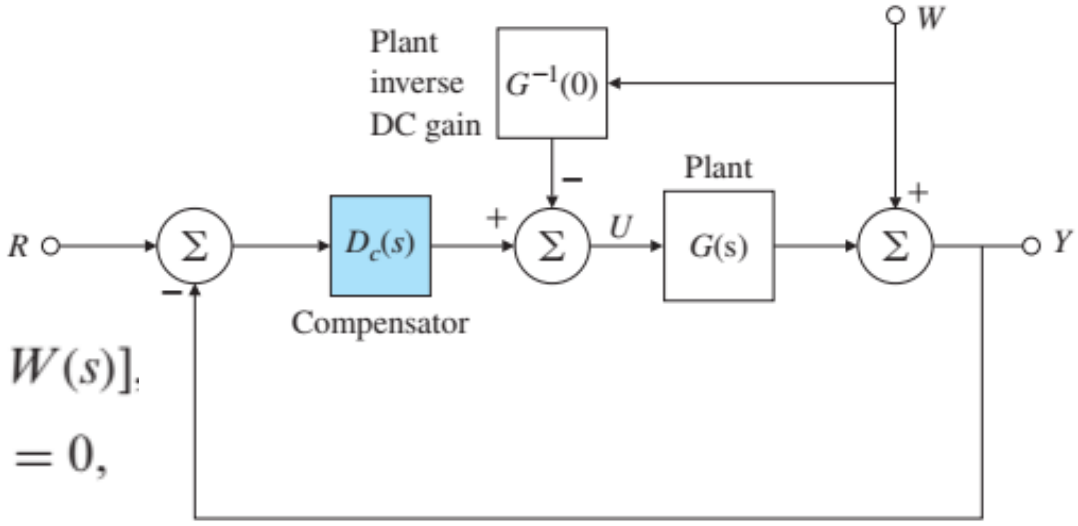
2. Feedforward Control by Plant Model Inversion

Example: Feedforward Control for DC Motor

(b) Disturbance rejection

- DC gain is $G^{-1}(0) = 1$
- Dc(s) = kP

Structure for disturbance rejection



Closed-loop TF:

$$Y(s) = W(s) + G(s)[k_P E(s) - W(s)],$$

$$E(s) = R(s) - Y(s), \text{ with } R(s) = 0,$$

$$\frac{Y(s)}{W(s)} = \mathcal{T}_w(s) = \frac{1 - G(0)^{-1}}{1 + k_P G(s)}.$$

2. Feedforward Control by Plant Model Inversion

Example: Feedforward Control for DC Motor

(b) Disturbance rejection

